A FINITELY GENERATED SIMPLE NON-MF GROUP

SAUERS

A countable group is MF if it embeds in the unitary group of a norm matrix corona. We construct an explicit finitely generated simple group with property (T) that is not MF, answering negatively the longstanding question of whether every countable group is MF. Namely, we prove that

$$H = \mathrm{EL}_{12}\big(L_{\mathbb{F}_2}(1,2)\big)$$

has no nontrivial homomorphism to any MF group. The proof uses an explicit Whitehead matrix that compresses a Kazhdan subgroup into itself. One-sided Kazhdan transport makes a distinguished involution invisible in normalized Hilbert–Schmidt norm in every operator-norm asymptotic representation. A finite-order invariant-support reduction then converts this tracial collapse into an operator-norm obstruction. Simplicity propagates the obstruction to the whole group. The same example also has no nontrivial finite-dimensional linear representation over any field.

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1. Introduction

A countable group is *MF* if it embeds in the unitary group of a norm matrix corona

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \longrightarrow 0\}$$

is the c_0-direct sum. Equivalently, an MF group admits finite-dimensional unitary models whose multiplicative defects tend to zero in operator norm and which asymptotically separate its nonidentity elements. Throughout, *MF group* means this operator-norm notion of Carrión–Dadarlat–Eckhardt and Korchagin [CDE, Kor]. A stronger regular-norm approximation property is also called MF in parts of the recent literature [GKEMP]; it is not the convention used here.

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For a countable group G, define its MF radical by

$$\mathrm{Rad}_{\mathrm{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi\colon G\to\mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker\pi.$$

The group G is MF precisely when $\mathrm{Rad}_{\mathrm{MF}}(G)$ is trivial. If $\mathrm{Rad}_{\mathrm{MF}}(G) = G$, every homomorphism from G to an MF group is trivial.

More generally, for $N \trianglelefteq G$ define its *MF closure* by

$$\mathrm{cl}^G_{\mathrm{MF}}(N) = \bigcap\{\ker f : N \le \ker f,\ f\colon G \to M,\ M \text{ MF}\}.$$

Thus $\mathrm{Rad}_{\mathrm{MF}}(G) = \mathrm{cl}^G_{\mathrm{MF}}(1)$, and G/N is MF precisely when $\mathrm{cl}^G_{\mathrm{MF}}(N) = N$.

Proposition 1.1 (MF residual calculus)**.** *For every countable group G, the subgroup $\mathrm{Rad}_{\mathrm{MF}}(G)$ is fully invariant, the quotient $G/\mathrm{Rad}_{\mathrm{MF}}(G)$ is MF, and*

$$G/N \text{ is MF} \iff \mathrm{cl}^G_{\mathrm{MF}}(N) = N \qquad (N \trianglelefteq G).$$

In particular, G is MF if and only if its MF radical is trivial.

Proof. An intersection of kernels is normal. If $f\colon G \to K$ is a homomorphism and $x \in \mathrm{Rad}_{\mathrm{MF}}(G)$, then every corona homomorphism of K, composed with f, kills x; hence $f(x) \in \mathrm{Rad}_{\mathrm{MF}}(K)$. This proves functoriality and, for endomorphisms, full invariance.

Put $R = \mathrm{Rad}_{\mathrm{MF}}(G)$ and enumerate the nonidentity elements of G/R. For each one, the definition of R supplies a corona homomorphism that detects a representative. A diagonal direct sum of coordinate models produces one operator-norm asymptotic representation that detects them all: at stage n, take the direct sum of sufficiently late coordinates of the first n models, chosen so that the first n multiplication defects are at most $1/n$ and each of the first n marked elements retains a fixed positive operator-norm separation on its designated summand. Its corona homomorphism is faithful on G/R. Thus G/R is MF.

Applied to G/N, the same argument says that G/N is MF exactly when the intersection of the kernels of all MF-target maps killing N is N, which is the asserted closure criterion. □

The existence of a countable non-MF group was open [Sh]. Theorem A answers this question with an explicit simple Kazhdan group. For a nontrivial simple group the MF radical is either trivial or the whole group, so full MF radical is equivalent here to failure of MF.

Let $L_{\mathbb{F}_2}(1,2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{1}$$

For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem A. *Put $R = L_{\mathbb{F}_2}(1,2)$ and $H = \mathrm{EL}_{12}(R)$. Then H is nontrivial, simple, finitely generated, and has property* (T)*, and*

$$\mathrm{Rad}_{\mathrm{MF}}(H) = H.$$

Equivalently, every homomorphism from H to an MF group is trivial. In particular, H is non-MF.

Suppose that $L \le G$ has property (T), that $u \in G$ satisfies $uLu^{-1} \le L$, and that c centralizes L. For every operator-norm asymptotic representation (V_n) of G,

$$\left\| V_n([ucu^{-1}, \ell]) - 1 \right\|_2 \longrightarrow 0 \qquad (\ell \in L). \tag{2}$$

Inside H, an explicit element τ satisfies $\tau L \tau^{-1} \le L$ for the upper-left subgroup $L \cong \mathrm{EL}_3(R)$. The element $c = e_{34}(1)$ centralizes L, and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The element d is nontrivial and H is simple, so d normally generates H. Since the coefficient ring has characteristic two, d is an involution. The finite-order invariant-support theorem then gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$. Thus property (T) is used for the compressed subgroup L in the transport step; the passage from the Hilbert–Schmidt obstruction to the MF radical uses only the finite order and normal generation of d.

For an exact finite-dimensional representation, conjugation by a compressor defines an injective endomorphism of the commutant of L. This endomorphism is surjective because the commutant is finite-dimensional. Consequently, every finite-dimensional linear representation of H over every field is trivial.

The argument is specific to operator-norm approximation. It does not prove that H is nonsofic or nonhyperlinear: normalized Hilbert–Schmidt approximations do not provide the operator-norm control used to construct the conjugation representation in Theorem 3.4.

The group H also gives non-MF groups with prescribed MF quotients. The following theorem is proved in Section 7.

Theorem B (prescribed MF quotients)**.** *For every countable group Q, there are a countable group W_Q and a split epimorphism $\pi_Q \colon W_Q \to Q$ such that*

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

The group W_Q is non-MF. If Q is MF, then

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\!\langle d \rangle\!\rangle_{W_Q},$$

where d is the element in Proposition 6.2. Moreover, precomposition with π_Q induces a bijection

$$\mathrm{Hom}(Q, M) \longrightarrow \mathrm{Hom}(W_Q, M)$$

for every MF group M.

Relation to prior work. The group-MF framework used here is developed by Carrión–Dadarlat–Eckhardt and Korchagin [CDE, Kor]; recent permanence results are due to Shulman [Sh]. The operator-norm case remained open after non-approximability had been established for the finite Schatten norms. Bachner–Dogon–Lubotzky emphasize the interaction between operator-norm defects and Hilbert–Schmidt rigidity as a possible route to the MF problem [BDL]. Our mechanism uses that interaction differently: one-sided compression of a Kazhdan subgroup forces Hilbert–Schmidt collapse, and the finite-order invariant-support reduction discards the common trivial summand of the resulting finite-dimensional models.

Leavitt introduced the algebras $L_K(1,n)$ as examples of rings with nonunique module rank [Lev]. Their defining relations supply the internal compression used here. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser's normal-subgroup theorem implies simplicity of the elementary group [Pre], while Ershov–Jaikin-Zapirain give property (T) for $\mathrm{EL}_n(R)$ whenever $n \geq 3$ and R is finitely generated [EJZ]. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

2. One-sided compression in finite dimension

Theorem 2.1 (finite-dimensional commutant rigidity)**.** *Let G be a group, let $L \leq G$, and suppose $u \in G$ satisfies $uLu^{-1} \leq L$. If k is a field, V is a finite-dimensional k-vector space, and $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\mathrm{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \qquad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned}\rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u)\\ &= \rho(u)^{-1}x\rho(h')\rho(u)\\ &= \rho(u)^{-1}x\rho(u)\rho(h).\end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. It commutes with every $\rho(\ell)$, $\ell \in L$, which proves the last assertion. □

3. Kazhdan transport in normalized Hilbert–Schmidt norm

For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator-norm asymptotic representation* of a countable group G is a sequence of maps $V_n\colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \longrightarrow 0\}.$$

The operator-norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G. Define

$$R_{\infty\to 2}(G) = \bigcap_V K_2(V), \tag{3}$$

where V ranges over all operator-norm asymptotic representations of G. An element is universally Hilbert–Schmidt trivial precisely when it belongs to $R_{\infty\to 2}(G)$.

Theorem 3.1 (finite-order normal generators)**.** *Let G be finitely generated and let $a \in G$ have finite order and normally generate G. Then*

$$a \in \mathrm{Rad}_{\mathrm{MF}}(G) \quad \iff \quad a \in R_{\infty\to 2}(G).$$

Consequently, if a is universally Hilbert–Schmidt trivial, then $\mathrm{Rad}_{\mathrm{MF}}(G) = G$.

Lemma 3.2. *For every sequence (m_n), the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v. Then $x_n^*x_n \to 1$ in norm. For all sufficiently large n, the matrix x_n is invertible, and the unitary $x_n(x_n^*x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. In a stably finite unital C^*-algebra, equivalent projections $p \le q$ satisfy $p = q$. □

Lemma 3.3 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C^*_{\max}(L) \to B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \le P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\operatorname{Fix}\pi(L)$. If ξ is L-fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^* \operatorname{Fix}\pi(L) \subseteq \operatorname{Fix}\pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B. □

Theorem 3.4 (one-sided Kazhdan transport)**.** *Let $L \leq G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator-norm asymptotic representation of G, and let (x_n) be uniformly bounded in operator norm. If*

$$\|V_n(\ell)x_nV_n(\ell)^* - x_n\|_2 \longrightarrow 0 \qquad (\ell \in L),$$

then, for every $\ell \in L$,

$$\|V_n(\ell)V_n(u)x_nV_n(u)^*V_n(\ell)^* - V_n(u)x_nV_n(u)^*\|_2 \longrightarrow 0,$$
$$\|V_n(\ell)V_n(u)^*x_nV_n(u)V_n(\ell)^* - V_n(u)^*x_nV_n(u)\|_2 \longrightarrow 0.$$

Proof. Let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the Hilbert-space ultraproduct $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \leq 2\,\|A - B\|\,,$$

so the operator-norm multiplicative defects of (V_n) also make the conjugation actions multiplicative modulo c_0. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\widetilde{\sigma}(g)$ be the class of the coordinate operators $\mathrm{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C}))/\bigoplus_n B(M_{d_n}(\mathbb{C})).$$

The preceding estimate shows directly that $g \mapsto \widetilde{\sigma}(g)$ is an exact homomorphism $G \to \mathcal{U}(\mathcal{B})$. Its restriction to L therefore extends to a $*$-homomorphism $C^*_{\max}(L) \to \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\operatorname{Fix}\sigma(L)$ [BHV, Chapter 3]. Since $uLu^{-1} \leq L$, the unitary $U = \widetilde{\sigma}(u)$ satisfies the hypothesis of Lemma 3.3, and hence $U^*PU \leq P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 3.2 therefore gives $U^*PU = P$. Hence both U and U^* preserve $\operatorname{Fix}\sigma(L)$, so $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. The two limits in the statement therefore vanish along ω. Since this holds for every free ultrafilter, both sequences converge ordinarily. □

Corollary 3.5. *Let $L \leq G$ have property* (T)*, let $uLu^{-1} \leq L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G) \qquad (\ell \in L).$$

Proof. Let (V_n) be an operator-norm asymptotic representation. If $c \in C_G(L)$, the sequence $(V_n(c))$ asymptotically commutes with $V_n(L)$ in operator norm and hence in normalized Hilbert–Schmidt norm. Theorem 3.4 shows that the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$. Therefore

$$\left\|V_n([ucu^{-1},\ell])-1\right\|_2 \longrightarrow 0 \qquad (\ell\in L).$$

Intersecting over (V_n) proves the claim. □

4. Finite-order invariant supports and the MF radical

The normalized Hilbert–Schmidt norm can lose a fixed operator-norm defect when the model contains a large trivial summand. For a finite-order normal generator, fixed normal words enlarge its nontrivial spectral subspace to an invariant representation space whose dimension is only a fixed multiple larger. Restricting to this invariant space simply deletes the trivial summand and produces another finite-dimensional unitary model.

Lemma 4.1 (unitary lifting). *Every unitary in $\mathcal{Q}_{\mathbf{d}}$ has a representative (U_n) with $U_n\in\mathcal{U}(d_n)$ for every n.*

Proof. Let (x_n) be a bounded lift of a corona unitary. Then $x_n^*x_n\to 1$ and $x_nx_n^*\to 1$ in operator norm. For all sufficiently large n, the matrix x_n is invertible and

$$U_n = x_n(x_n^*x_n)^{-1/2}$$

is unitary with $\|U_n-x_n\|\to 0$. Define the finitely many remaining coordinates arbitrarily. □

Lemma 4.2 (finite-spectrum rounding). *Let $U_n\in\mathcal{U}(d_n)$ and suppose $\|U_n^m-1\|\to 0$ for some $m\geq 2$. Then there are $B_n\in\mathcal{U}(d_n)$ such that*

$$B_n^m=1,\qquad \|B_n-U_n\|\longrightarrow 0.$$

Proof. Round each point of the spectrum of U_n to a nearest mth root of unity and apply the resulting function by finite-dimensional functional calculus. The rounded unitary has mth power one. Compactness of the unit circle gives

$$\sup\{\operatorname{dist}(z,\mu_m): |z|=1, |z^m-1|\leq\varepsilon\}\longrightarrow 0 \quad (\varepsilon\downarrow 0),$$

which proves the norm convergence. □

Theorem 4.3 (invariant-support reduction). *Let G be generated by a finite symmetric set S containing an element a of order $m\geq 2$, and suppose that a normally generates G. For each $s\in S$, fix an expression of s as a product of conjugates of $a^{\pm 1}$, taking the expression for a to be the one-letter word a. Let L be the total number of occurrences of $a^{\pm1}$ in these expressions.*

If $\rho\colon G\to\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ satisfies $\rho(a)\neq 1$, then, after passing to an infinite coordinate subsequence and changing the chosen representatives by $o(1)$ in operator norm, the same corona homomorphism has representatives of the form

$$\sigma_n(g)\oplus 1_{d_n-k_n},\qquad \sigma_n(g)\in\mathcal{U}(k_n),$$

where (σ_n) *is an operator-norm asymptotic representation. Moreover,* $A_n = \sigma_n(a)$ *satisfies* $A_n^m = 1$*, and if*

$$p_n = 1_{\mathbb{C}\setminus\{1\}}(A_n), \qquad r_n = \operatorname{rank} p_n,$$

then

$$1 \le r_n \le k_n \le L r_n. \tag{4}$$

Consequently, with

$$\eta_m = \min\{|1-\zeta| : \zeta^m = 1,\ \zeta \neq 1\} > 0,$$

one has

$$\|\sigma_n(a) - 1\|_2 \ge \frac{\eta_m}{\sqrt{L}}. \tag{5}$$

If a *is a product of* N_g *conjugates of* $g^{\pm 1}$*, then also*

$$\liminf_n \|\sigma_n(g) - 1\|_2 \ge \frac{\eta_m}{N_g\sqrt{L}}. \tag{6}$$

Proof. By Lemma 4.1, choose unitary lifts $U_n(g)$ of $\rho(g)$. They form an operator-norm asymptotic representation. Since $\rho(a)^m = 1$, Lemma 4.2 replaces $U_n(a)$, within $o(1)$ in operator norm, by a unitary B_n satisfying $B_n^m = 1$.

Write the chosen expression for $s \in S$ as

$$s = \prod_{j=1}^{L_s} h_{s,j} a^{\varepsilon_{s,j}} h_{s,j}^{-1}, \qquad \varepsilon_{s,j} \in \{1, -1\}.$$

Define

$$\widetilde{U}_n(s) = \prod_{j=1}^{L_s} U_n(h_{s,j}) B_n^{\varepsilon_{s,j}} U_n(h_{s,j})^*.$$

For every $g \in G$, choose once and for all a word in S representing g and define $\widetilde{U}_n(g)$ by substituting these matrices. We use the one-letter word for a, so $\widetilde{U}_n(a) = B_n$. The defining group identities and asymptotic multiplicativity give

$$\left\|\widetilde{U}_n(g) - U_n(g)\right\| \longrightarrow 0 \qquad (g \in G).$$

Thus $(\widetilde{U}_n)$ is an operator-norm asymptotic representation and represents the same corona homomorphism.

Let P_n be the active spectral subspace of B_n, and let K_n be the sum of the subspaces

$$U_n(h_{s,j}) P_n \qquad (s \in S,\ 1 \le j \le L_s).$$

Every factor in the displayed formula for $\widetilde{U}_n(s)$ is the identity on $K_n^\perp$, preserves K_n, and differs from the identity by an operator whose range lies in K_n. Hence every $\widetilde{U}_n(g)$ preserves K_n and is the identity on $K_n^\perp$. Since the one-letter expression for a occurs in the family,

$$\dim P_n \le \dim K_n \le L \dim P_n.$$

The assumption $\rho(a) \neq 1$ makes P_n nonzero on an infinite coordinate subsequence. Retain that subsequence, put $k_n = \dim K_n$, and let $\sigma_n(g)$ be the restriction of $\widetilde{U}_n(g)$ to K_n. This proves the lifting assertion and (4).

Every nonidentity eigenvalue of A_n is an mth root of unity, so

$$\|A_n - 1\|_2^2 \geq \eta_m^2 \frac{r_n}{k_n} \geq \frac{\eta_m^2}{L},$$

which proves (5). This is the ordinary normalized Hilbert–Schmidt norm on $M_{k_n}(\mathbb{C})$, applied to the restricted unitary representation. Finally, substitute the matrices $\sigma_n(g)$ into a fixed normal word of length N_g expressing a through $g^{\pm 1}$. Telescoping in normalized Hilbert–Schmidt norm and asymptotic multiplicativity give

$$\|\sigma_n(a) - 1\|_2 \leq N_g \, \|\sigma_n(g) - 1\|_2 + o(1).$$

Together with (5), this proves (6). □

Proof of Theorem 3.1. For every group,

$$\mathrm{Rad}_{\mathrm{MF}}(G) \leq R_{\infty\to 2}(G):$$

an operator-norm asymptotic representation defines a homomorphism to its norm matrix corona, and an element killed there converges to the identity in operator norm, hence in normalized Hilbert–Schmidt norm.

Conversely, suppose $a \in R_{\infty\to 2}(G)$ but $a \notin \mathrm{Rad}_{\mathrm{MF}}(G)$. Some corona homomorphism sees a. Theorem 4.3 produces an operator-norm asymptotic representation for which (5) holds, contradicting the definition of $R_{\infty\to 2}(G)$. Thus a lies in the MF radical. Since that radical is normal and a normally generates G, it equals G. □

Corollary 4.4 (simple torsion dichotomy)**.** *Let G be a nontrivial finitely generated simple group containing a finite-order nonidentity element. Then*

$$\mathrm{Rad}_{\mathrm{MF}}(G) = R_{\infty\to 2}(G).$$

Exactly one of the following holds:

(1) every homomorphism from G to a norm matrix corona is trivial;
(2) G admits a faithful operator-norm asymptotic representation (σ_n) such that

$$\liminf_n \|\sigma_n(g) - 1\|_2 > 0 \qquad (g \neq 1).$$

Proof. Both displayed residuals are normal subgroups. Simplicity and Theorem 3.1, applied to a nonidentity torsion element, give their equality. If a corona homomorphism is nontrivial, it is injective. Apply Theorem 4.3; for each $g \neq 1$, simplicity puts the chosen torsion element in the normal closure of g, so (6) gives the asserted pointwise separation. The two alternatives are therefore exhaustive and mutually exclusive. □

5. The binary Leavitt self-compression

Throughout this section $R = L_{\mathbb{F}_2}(1,2)$ and

$$p = s_0 t_0, \qquad q = s_1 t_1.$$

The relations (1) give

$$p + q = 1, \qquad t_1 q s_1 = 1. \tag{7}$$

In particular $q \neq 0$.

We use indices $0, \dots, 11$ for 12×12 matrices and identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ consisting of the matrices $\mathrm{diag}(A, I_9)$. On $M_3(R)$ consider the unital injective endomorphism

$$\Psi(A) = qI_3 + s_0 A t_0. \tag{8}$$

Here $s_0 A t_0$ denotes the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $qs_0 = t_0 q = 0$ and $t_0 s_0 = 1$ show that Ψ is multiplicative and unital. The identity $t_0 \Psi(A) s_0 = A$ proves injectivity. The term qI_3 is needed for unitality because the map $A \mapsto s_0 A t_0$ sends I_3 to pI_3.

Set

$$U = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \qquad V = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix}.$$

A direct calculation using (1) gives $UV = VU = I_6$. Put

$$\tau = \mathrm{diag}(U, V) \in \mathrm{GL}_{12}(R). \tag{9}$$

Writing $I = I_6$, the Whitehead factorization is

$$\mathrm{diag}(U, U^{-1}) = \begin{pmatrix} I & U \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -U^{-1} & I \end{pmatrix} \begin{pmatrix} I & U \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}. \tag{10}$$

Each block-unipotent factor is a product of elementary 12×12 matrices: its off-diagonal 6×6 block may be inserted one entry at a time, and the corresponding matrix units have pairwise zero products. Since $V = U^{-1}$, (10) proves $\tau \in \mathrm{EL}_{12}(R)$.

Let $H = \mathrm{EL}_{12}(R)$ and let $L = \mathrm{EL}_3(R) \leq H$ be the upper-left corner. Both groups have property (T) by the theorem of Ershov and Jaikin-Zapirain [EJZ]. Block multiplication yields

$$\tau \,\mathrm{diag}(A, I_9)\, \tau^{-1} = \mathrm{diag}(\Psi(A), I_9). \tag{11}$$

For every $i \neq j$ and $a \in R$, $\Psi(e_{ij}(a)) = e_{ij}(s_0 a t_0)$. Therefore

$$\tau L \tau^{-1} \leq L. \tag{12}$$

6. The full MF radical

Proposition 6.1. *The group $H = \mathrm{EL}_{12}(R)$ is simple.*

Proof. The ring $R = L_{\mathbb{F}_2}(1,2)$ is purely infinite simple [AA] and therefore an exchange ring [Ara]. Let $N \trianglelefteq H$. Preusser's normal-subgroup theorem [Pre, Theorem 3] gives an ideal $I \trianglelefteq R$ such that

$$\mathrm{EL}_{12}(R,I) \leq N \leq C_{12}(R,I),$$

where $C_{12}(R,I)$ is the full congruence subgroup of level I. Since R is simple, either $I = R$ or $I = 0$. In the first case $\mathrm{EL}_{12}(R,I) = H$, so $N = H$. In the second case

$$N \leq C_{12}(R,0) = Z(\mathrm{GL}_{12}(R)).$$

The center of R is $\mathbb{F}_2$ by [AC, Corollary 4.3]. For completeness, suppose that $z \in Z(\mathrm{GL}_{12}(R))$. Commuting z with the elementary matrices $e_{ij}(1)$ shows that $z = \lambda I_{12}$ for some $\lambda \in R$. Commuting with $e_{ij}(a)$ for arbitrary $a \in R$ then gives $\lambda a = a\lambda$, so $\lambda \in Z(R)^\times = \mathbb{F}_2^\times = \{1\}$. Thus $Z(\mathrm{GL}_{12}(R))$ is trivial, and $N = 1$ when $I = 0$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. $\square$

Proposition 6.2. *Let*

$$c = e_{34}(1), \qquad \ell = e_{12}(1), \qquad d = [\tau c \tau^{-1}, \ell]$$

in H. Then

$$c \in C_H(L), \qquad d = e_{02}(q) \neq 1, \qquad d^2 = 1, \qquad \langle\!\langle d \rangle\!\rangle_H = H.$$

Proof. Every elementary generator of the upper-left 3×3 corner has both indices in $\{0,1,2\}$. The elementary commutator relations show that it commutes with $c = e_{34}(1)$, so $c \in C_H(L)$.

Direct multiplication of the matrices in (9) gives

$$\tau c \tau^{-1} = e_{01}(q) e_{34}(1). \tag{13}$$

The second factor commutes with $\ell = e_{12}(1)$, and $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ gives $d = e_{02}(q)$. This is nontrivial because $q \neq 0$. Since R has characteristic two and $E_{02}^2 = 0$, one also has $e_{02}(q)^2 = 1$. Finally, d is nontrivial and H is simple by Proposition 6.1, so its normal closure is all of H. $\square$

Proof of Theorem A. The group H is nontrivial and simple by Proposition 6.1. The ring R is finitely generated, and H has property (T) by the Ershov–Jaikin-Zapirain theorem for elementary groups over finitely generated rings [EJZ]; in particular H is finitely generated.

The containment (12), the centrality of c relative to L, and Corollary 3.5 give $d \in R_{\infty\to 2}(H)$. The subgroup $R_{\infty\to 2}(H)$ is normal, and Proposition 6.2 says that d is an involution which normally generates H. Theorem 3.1 gives $d \in \mathrm{Rad}_{\mathrm{MF}}(H)$. Normality of the MF radical now gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$.

If M is MF and $f\colon H \to M$ is a homomorphism, compose f with a faithful homomorphism from M to a norm matrix corona. Since $\mathrm{Rad}_{\mathrm{MF}}(H) = H$, this composition and hence f are trivial. If H were MF, its identity homomorphism would contradict this conclusion. $\square$

Corollary 6.3. *For every field k and every finite-dimensional k-vector space V, every homomorphism $H \to \mathrm{GL}(V)$ is trivial.*

Proof. Let $\rho\colon H \to \mathrm{GL}(V)$. The containment (12), the centrality of c relative to L, and Theorem 2.1 give $\rho(d) = 1$. Since $H = \langle\!\langle d \rangle\!\rangle_H$ by Proposition 6.2, the homomorphism ρ is trivial. □

In particular, every homomorphism from H to a finite group, a residually finite group, or a compact Hausdorff group is trivial. For finite and residually finite targets this follows by the regular representation of a finite quotient; for compact targets it follows from Peter–Weyl point separation [PW].

Corollary 6.4 (unit groups and all matrix ranks)**.** *Put $R = L_{\mathbb{F}_2}(1,2)$. Then the unit group $R^\times$ and $\mathrm{GL}_n(R)$ for every $n \geq 1$ are non-MF.*

Proof. For $n \geq 2$, the complete binary prefix code

$$\mathcal{C}_n = \{1^j 0 : 0 \leq j \leq n-2\} \cup \{1^{n-1}\}$$

(and $\mathcal{C}_1$ consisting of the empty word) gives a ring isomorphism

$$\Theta_n\colon M_n(R) \longrightarrow R, \qquad (a_{uv}) \longmapsto \sum_{u,v \in \mathcal{C}_n} s_u a_{uv} t_v. \tag{14}$$

Here $s_{i_1\cdots i_r} = s_{i_1} \cdots s_{i_r}$ and $t_{i_1\cdots i_r} = t_{i_r} \cdots t_{i_1}$. Indeed,

$$t_u s_v = \delta_{uv}, \qquad \sum_{u \in \mathcal{C}_n} s_u t_u = 1,$$

so the inverse of (14) is $r \mapsto (t_u r s_v)_{u,v}$.

Taking units gives $\mathrm{GL}_n(R) \cong R^\times$ for every n. For $n = 12$, the subgroup $H = \mathrm{EL}_{12}(R) \leq \mathrm{GL}_{12}(R)$ therefore embeds in $R^\times$. If $R^\times$ were MF, then its subgroup H would be MF, contrary to Theorem A. Hence $R^\times$ is non-MF, and the displayed isomorphisms give the result for every $\mathrm{GL}_n(R)$. □

7. Prescribed MF quotients

Let $A = \langle d \rangle \leq H$, where d is the element in Proposition 6.2. For a countable group Q, define

$$W_Q = H *_A (Q \times A), \tag{15}$$

where $A \to Q \times A$ is the inclusion into the second factor. The maps from H and $Q \times A$ to Q that are respectively trivial and the projection onto Q agree on A. They therefore define an epimorphism

$$\pi_Q\colon W_Q \longrightarrow Q.$$

It is split by the inclusion $Q \to Q \times A \to W_Q$. The normal-form theorem for amalgamated free products shows that both vertex maps in (15) are injective; in particular, $d \neq 1$ in W_Q.

Proposition 7.1 (universal factorization)**.** *Let T be a group such that every homomorphism $H \to T$ is trivial. Then precomposition with π_Q is a bijection*

$$\mathrm{Hom}(Q, T) \longrightarrow \mathrm{Hom}(W_Q, T).$$

Furthermore,

$$\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

Proof. Let $f \colon W_Q \to T$ be a homomorphism. Its restriction to H is trivial, so its restriction to the amalgamated subgroup A is trivial. For $(q, a) \in Q \times A$, we then have

$$f(q, a) = f(q, 1)f(1, a) = f(q, 1).$$

Thus the restriction of f to $Q \times A$ factors uniquely through the projection onto Q. The universal property of the amalgam shows that f factors through π_Q. The factorization is unique because π_Q is surjective.

The element d belongs to $\ker \pi_Q$. Conversely, quotienting W_Q by $\langle\langle d \rangle\rangle_{W_Q}$ kills H, since $\langle\langle d \rangle\rangle_H = H$, and kills the second factor A in $Q \times A$. The resulting quotient is Q, with quotient map induced by π_Q. Hence $\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}$. □

Proof of Theorem B. Every homomorphism from H to an MF group is trivial by Theorem A. Proposition 7.1 therefore gives the asserted bijection for every MF group M. In particular, every homomorphism from W_Q to a norm matrix corona factors uniquely through π_Q. Intersecting their kernels gives

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

The kernel of π_Q belongs to this radical, and it contains the nonidentity element d. Thus W_Q is non-MF. If Q is MF, its MF radical is trivial, and the displayed equality becomes

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

□

The same factorization applies to every target group T for which all homomorphisms $H \to T$ are trivial. Corollary 6.3 and Peter–Weyl point separation therefore give the corresponding bijections for finite-dimensional linear groups over arbitrary fields, residually finite groups, and compact Hausdorff groups.

AI and computational resource usage

AI systems were used extensively for proof exploration, literature navigation, formalization, and editorial revision. Selected supporting lemmas and related variants have been formalized in Lean; this does not constitute an end-to-end formal verification of the present manuscript. The author checked and revised the mathematical arguments and citations.

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